Forward College	**2, LEBUH ACHEH, GEORGE TOWN** **10300 GEORGE TOWN** **PULAU PINANG** **PENILAIAN AKHIR TEORI**	
PROGRAM'S CODE & NAME	J620-002-4:2020 FRONT-END SOFTWARE DEVELOPMENT	
LEVEL	4 (FOUR)	
COMPETENCY UNIT NO. AND TITLE	J620-002-4:2020-C01: RESPONSIVE WEBSITE DEVELOPMENT	
WORK ACTIVITIES NO. AND STATEMENT	**1. CREATE WEB DESIGN MOCK-UP ELEMENTS.** **2. PLAN WEB DESIGN STRUCTURE.** **3. TRANSFORM MOCK-UP TO RESPONSIVE WEB PAGES.** **4. VERIFY DEVELOPED RESPONSIVE WEBSITE.** **5. VERIFY WEBSITE ACCESSIBLE GLOBALLY.**	
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NAME		
IC NUMBER		
DATE		
DURATION	**2.5 hours**	**MARKS: / 100%**
SIGNATURE OF PEGAWAI PENILAI		
KERTAS PEPERIKSAAN INI DISAHKAN OLEH JAWATANKUASA KELULUSAN SOALAN		

Instruction for the candidates:

1. Write your name and IC number in the space provided.
2. You shall only start reading the questions when you are allowed to do so.
2. Answer all questions.
3. No notes were allowed in the examination hall unless informed to do so.
4. Cheating is prohibited.
5. You are not allowed to bring this question paper out from this hall.

THIS PAPER CONSISTS OF 24 PRINTED PAGES

SECTION A: MULTIPLE CHOICE QUESTIONS

Instruction: Choose the correct answer. Write your answer beside the number. (1 point for every correct answer)

1. Select the proper representation of the HTML tag <h1>.
    A. A link
    B. An important heading
    C. A paragraph
    D. An italic text

2. Choose the HTML tag that is used to create a paragraph.
    A. <p>
    B.
    C.
    D. <h3>

3. Choose the correct purpose of the tag in HTML?
    A. To create a link
    B. To make the text italic
    C. To make text bold
    D. To create a paragraph

4. Select the language that controls a webpage's layout, colours, fonts, and other visual elements.
    A. JavaScript
    B. HTML
    C. CSS
    D. PHP

5. Choose the property that is used to enable the Flexbox layout in CSS.
    A. flex-container
    B. flex-display
    C. display
    D. flex-layout

6. Select the default value of the flex-direction property in a Flexbox layout.
    A. row
    B. row-reverse
    C. column
    D. column-reverse

7. Choose the property that is used to create a grid container in CSS
    A. display: block;
    B. display: flex;
    C. display: grid;
    D. display: inline;

8. Choose the correct syntax to create a class selector in CSS.
    A. .classname
    B. #classname
    C. classname
    D. *classname

9. Choose the correct symbol used to create an ID selector in CSS
    A. .
    B. #
    C. *
    D. %

10. Select the purpose of Google Web Fonts
    A. To optimize website performance
    B. To provide a library of free fonts for web projects
    C. To validate HTML and CSS code
    D. To create custom animations

11. Select the property used to control the stacking order of positioned elements on a webpage.
    A. order
    B. stack
    C. z-index
    D. z-order

12. Select the type of positioning used to create elements that stick to a certain position as the user scrolls the page and then "unsticks" once the element reaches a certain point in the viewport
    A. Fixed positioning
    B. Absolute positioning
    C. Relative positioning
    D. Sticky positioning

13. Choose the total number of columns in the Bootstrap grid system
    A. 6
    B. 10
    C. 12
    D. 16

14. Select the correct class that set the font size of an element to the largest size using Bootstrap's text utilities
    A. .fs-1
    B. .fs-3
    C. .fs-6
    D. .fs-max

15. Select the data type used to represent true or false values in JavaScript.
    A. String
    B. Number
    C. Boolean
    D. Object

16. Select the starting index for elements in an array in JavaScript.
    A. -1
    B. 0
    C. 1
    D. 2

17. Choose the correct method to add an element to the end of an array.
    A. push()
    B. pop()
    C. shift()
    D. unshift()

18. Choose the correct media feature rule that targets screens 768 pixels wide or narrower.
    A. max-width: 768px
    B. min-width: 768px
    C. width: 768px
    D. screen-width: 768px

19. Choose the correct purpose of adding a breakpoint to a column class in Bootstrap
    A. To change the color of column
    B. To make the column non-responsive
    C. To create a fixed-width column
    D. To make the column responsive at specific screen widths

20. Select the best answer which explains the main difference between unordered and ordered lists in HTML?
    A. Unordered lists use bullet points, while ordered lists use numbers.
    B. Unordered lists can only contain text, while ordered lists can contain images.
    C. Unordered lists can only have one level, while ordered lists can have multiple levels.
    D. Unordered lists use the <ul> tag, while ordered lists use the <li> tag.

21. Select the HTML tags to create a table row and data cell.
    A. <tr> and <td>
    B. <thead> and <tbody>
    C. <table> and <tr>
    D. <th> and <tfoot>

22. Select the difference between the space-between and space-around values of the justify-content property in Flexbox.
    A. space-between aligns elements at the container edges, while space-around aligns them at the center.
    B. space-between evenly distributes space between elements, while space-around distributes space around elements.
    C. space-between allows elements to wrap onto multiple lines, while space-around prevents wrapping.
    D. space-between distributes space in reverse order, while space-around distributes space in normal order.

23. Select the CSS property that can be used to position a grid item by defining its starting and ending grid lines for both columns and rows using shorthand notation
    A. grid-position
    B. grid-area
    C. grid-column and grid-row
    D. grid-start-end

24. Choose the correct method to create different gap sizes for rows and columns in a grid container.
    A. Use 'grid-gap' with two different values for row and column gaps
    B. Use 'grid-row-gap' for row gaps and 'grid-column-gap' for column gaps
    C. Set different padding values for the grid items
    D. Use 'grid-gap' with a single value, and then overwrite it with 'grid-row-gap' or 'grid-column-gap'

25. Choose the correct CSS selector that can be used to select all elements that are direct children of a specified element
    A. Descendant selector
    B. Child selector
    C. Adjacent sibling selector
    D. Attribute selector

26. Choose the correct CSS syntax that selects and styles all the heading elements (h1, h2, h3, h4, h5, h6) that have the class "title"
    A. h.title { ... }
    B. h1.title, h2.title, h3.title, h4.title, h5.title, h6.title { ... }
    C. .title h1, h2, h3, h4, h5, h6 { ... }
    D. h1.title > h2.title > h3.title > h4.title > h5.title > h6.title { ... }

27. Choose the correct CSS syntax that adjusts the font size of an element using Bootstrap text utility classes
    A. .font-size-N (where N is a number between 1 and 6)
    B. .fs-N (where N is a number between 1 and 6)
    C. .text-size-N (where N is a number between 1 and 6)
    D. .size-N (where N is a number between 1 and 6)

28. Choose the correct Bootstrap class that can be used to center align text within an element
    A. .text-left
    B. .text-right
    C. .text-center
    D. .text-middle

29. Choose the correct operator that is used for strict equality in JavaScript
    A. ==
    B. !=
    C. ===
    D. !==

30. Choose the output of the following code.

```
const fruits = ['apple', 'banana', 'orange'];
const result = fruits.slice(1, 3);
console.log(result);
```

    A. ['apple', 'banana']
    B. ['banana', 'orange']
    C. ['apple', 'orange']
    D. ['banana']

31. Select the primary advantage of using relative units of measurement like em and rem in typography compared to absolute units like pixels (px)
    A. They allow for more precise control over text size.
    B. They make text elements scale proportionally with the viewport size.
    C. They make the text appear more visually appealing.
    D. They allow for faster load times.

32. Select the main purpose of the <meta name="viewport" content="width=device-width, initial-scale=1.0" /> tag in an HTML document
    A. To set the character encoding for the webpage.
    B. To help older versions of Internet Explorer render the webpage correctly.
    C. To make the website responsive on different devices like smartphones, tablets, and computers.
    D. To define the title of the webpage.

33. Choose the main difference between block-level elements and inline elements in HTML.
    A. Block-level elements are used for styling, while inline elements are used for the structure.
    B. Block-level elements take up the full width of their container, while inline elements only take up as much width as necessary for their content.
    C. Inline elements always start on a new line, while block-level elements do not.
    D. Block-level elements can only be used within other block-level elements, while inline elements can be used within both block-level and inline elements.

34. Select the correct statement that accurately describes the flex-direction property values in Flexbox

    I. row - elements are placed from left to right
    II. row-reverse - elements are placed from right to left
    III. column - elements are placed from left to right
    IV. column-reverse - elements are placed from bottom to top

    A. I, II, III
    B. I, II, IV
    C. I, III, IV
    D. II, III, IV

35. Determine the correct descriptions of units of measurement in CSS for typography:

    I. Pixels (px) is an absolute unit of measurement that specifies the exact size of an element on the screen.
    II. Ems (em) is a relative unit of measurement that is based on the font size of the parent element.
    III. Rems (rem) is a relative unit of measurement based on the font size of the root element.
    IV. Kilograms (kg) is a CSS unit of measurement used for the font weight.

A. I, II, III
B. I, II, IV,
C. I, III, IV
D. II, III, IV

36. Choose the correct statements about the 'fr' unit in CSS Grid:

    I. 'fr' stands for "fractional unit."
    II. The 'fr' unit is used to divide the available space in the grid container evenly between the grid tracks.
    III. The 'fr' unit can be used to set the size of grid gaps.
    IV. The 'fr' unit can only be used with the 'grid-template-columns' property.

A. I, II, III,
B. I, II, IV,
C. II, III, IV
D. I, III, IV

37. Determine the correct descriptions of JavaScript data types:

    I. Strings are used to represent text and are enclosed in single or double quotes.

    II. Numbers are used for mathematical computations and can represent integers or floating-point values.

    III. Booleans are used to represent a collection of related data values.

    IV. Objects are key-value pairs, where keys are strings, and values can be any data type.

    A. I, II, III
    B. I, II, IV
    C. I, III, IV
    D. II, III, IV

38. Determine the correct descriptions of Media Queries:

    I. Media Queries provide a way to apply different styles to a webpage based on the characteristics of the device it's being viewed on.

    II. Media Queries can be used to adjust layout, typography, and other design elements based on the device being viewed on.

    III. Media Queries only work with HTML and are not compatible with CSS.

    IV. Media Queries have a basic syntax that looks like this: @media media-type and (media-feature-rule) { /* CSS rules go here */ }

    A. I, II, III
    B. I, II, IV
    C. I, III, IV
    D. II, III, IV

39. Determine the accurate statements about the following code:

```
let i = 1;
let sum = 0;
while (i <= 10) {
    sum += i;
    i++;
}
console.log(sum);
```

I. The code calculates the sum of the numbers from 1 to 10.
II. The output will be 55.
III. The loop will execute 9 times.
IV. The loop will execute 10 times.

A. I, II, III
B. I, II, IV
C. I, III, IV
D. II, III, IV

40. Determine the correct descriptions of the <!DOCTYPE html> declaration:

    I. The <!DOCTYPE html> declaration must be the first line of an HTML document.

    II. The <!DOCTYPE html> declaration informs the browser that the document is written in HTML5.

    III. The <!DOCTYPE html> declaration is used to define the character encoding for the webpage.

    IV. The <!DOCTYPE html> declaration is a self-closing tag and doesn't require a closing tag.

A. I, II, III
B. I, II, IV
C. I, III, IV
D. II, III, IV

SECTION B: SUBJECTIVE QUESTION

1. Create a basic webpage using HTML and CSS that includes simple formatting and styling.

   I. Create an HTML structure that includes a title heading (h1) and two paragraphs. The first paragraph should include a bolded word and the second paragraph should include an italicized word. Add a class called "important" to the bolded word in the first paragraph.

```
<!DOCTYPE html>
<html>
<head>
  <title>Simple Webpage</title>
</head>
<body>
```

<!-- Only change code below this line -->

<!-- Only change code above this line -->

```
</body>
</html>
```

(10 Marks)

II. Create an internal CSS stylesheet for the HTML structure. Apply the following styling:

a. Set the font family for the entire document to "Arial, sans-serif".

b. Set the font size for the h1 element to 24px and the font color to blue.

```
<!DOCTYPE html>
<html>
<head>
  <title>Simple Webpage</title>
  <style>
    /* Only change code below this line */




    /* Only change code above this line */
  </style>
</head>
<body>
  <!-- (HTML structure from Part I) -->
</body>
</html>
```

(6 Marks)

III. Using the class selector, Set the font color to red for the "important" class:

```
<!DOCTYPE html>
<html>
<head>
  <title>Simple Webpage</title>
  <style>
    /* (CSS styling from Part II) */
    /* Only change code below this line */




    /* Only change code above this line */
  </style>
</head>
<body>
  <!-- (HTML structure from Part I) -->
</body>
</html>
```

(4 Marks)

2. Explain the basic Usage of Bootstrap Components

I. Using Bootstrap's grid system, create a row with three equal columns. In each column, place a heading (h3) with any text you prefer.

(6 Marks)

II. Explain how you would implement and use each of the following Bootstrap components in a web project:

a. A button with a secondary color

(4 Marks)

III) Design the HTML using CSS. Given a button with an id of `submit-btn`, target this id and apply the following styles:

- Set the background to green.
- Change the text color to white.
- Adjust the size of the font to 16px.
- Make the corners rounded with a radius of 5px.
- Add spacing inside the button with padding of 10px 20px.

//Write your code below this line

(5 marks)

IV) Write the HTML for a navigation bar with the following requirements:

It contains three links: Home, About, and Contact.
Use an unordered list (`<ul>`) to structure the links.

- Home
- About
- Contact

//Write your code below this line

(5 marks)

3. Implement a simple program using loops and arrays in JavaScript

I. Create an array called `fruits` containing the following strings: "apple", "banana", "orange", and "grape".

(4 Marks)

II. Using a `for` loop, iterate over the `fruits` array and print each element (using console.log) in the following format: "Fruit [index]: [fruit name]". For example, the first line should read "Fruit 1: apple".

(4 Marks)

III. Modifying the array using the following array methods

a. Add the string "strawberry" to the end of the `fruits` array using the appropriate array method.

(3 Marks)

b. Remove the first element ("apple") from the `fruits` array using the appropriate array method.

(3 Marks)

c. Insert the string "kiwi" at the beginning of the `fruits` array using the appropriate array method.

(3 Marks)

d. Using `slice()`, extract the second and third elements of the modified `fruits` array and print the result

(3 Marks)